



JAAFR
INTERNATIONAL
RESEARCH JOURNAL

JOURNAL OF ADVANCE AND FUTURE RESEARCH

JAAFR.ORG | ISSN : 2984-889X

An International Open Access, Peer-reviewed, Refereed Journal

The New Underwriting Trinity: A Comparative Analysis of UPI-Centric Credit Scoring Models in India

Shivam Gogia, Shiv Nandan Singh

Department of Data Science & Artificial Intelligence

School of Computer Applications, Babu Banarasi Das University, Lucknow, India

Abstract

The proliferation of India's Unified Payments Interface (UPI) has generated an unprecedented, high-velocity stream of transactional data, offering a powerful alternative to traditional credit bureau information. This has catalyzed the emergence of new credit scoring paradigms aimed at serving the millions of "thin-file" and "new-to-credit" individuals previously invisible to the formal financial system. This thesis, "The New Underwriting Trinity," conducts a comprehensive comparative analysis of three dominant UPI-centric credit scoring philosophies, synthesized from a systematic review of publicly available academic research, industry reports, and news articles. The three models examined are: the real-time Behavioral Scoring model, the historical Bank Statement Analysis (BSA) model, and the forward-looking Short-Term Rolling Features model designed for revolving credit. The analysis evaluates these conceptual frameworks across four critical dimensions: theoretical predictive power, model explainability, algorithmic fairness, and operational viability. The findings reveal a landscape of strategic trade-offs, where no single approach is universally superior. The Behavioral model offers high accuracy for detecting acute risk but carries significant operational complexity, high privacy risk, and low interpretability. The BSA model provides robust, stable, and interpretable assessments but is less sensitive to real-time changes in risk. The Rolling Features model is a specialized tool, purpose-built for the dynamic risk management of revolving credit products like Credit Line on UPI (CLOU). This thesis concludes that the optimal strategy for lenders is not to choose one model but to develop a nuanced, hybrid approach. This "stacked" methodology would leverage the stable, interpretable BSA model for foundational underwriting and setting initial credit limits, while employing the acute, real-time Behavioral or Rolling models for ongoing fraud detection, portfolio monitoring, and dynamic risk management. TABLE OF CONTENTS

Keywords: UPI, Alternative Credit Scoring, Behavioral Scoring, Bank Statement Analysis (BSA), Short-Term Rolling Features, CLOU, Cash-Flow Underwriting, Explainable AI (XAI), Algorithmic Fairness, Model Drift, RBI Guidelines, Financial Inclusion, India, Purchasing Power Parity (PPP), Cost-of-Living Normalization

I. INTRODUCTION

Introduction 1.1 The New Digital Financial Epoch in India India is currently in the midst of a profound financial transformation, arguably one of the most rapid and large-scale digital revolutions in modern history. This is not merely an incremental shift but a fundamental restructuring of how a nation of 1.4 billion people accesses, manages, and moves money. Driven by a confluence of enabling government policy, widespread mobile internet penetration, and a vibrant startup ecosystem, the country's financial landscape is being rebuilt on a foundation of digital public infrastructure. This new epoch is characterized by a "cash-to-digital" migration that is creating unprecedented efficiencies, new economic opportunities, and, most critically, a data-rich environment that holds the key to solving the nation's most enduring economic challenges.

1.2 The UPI Catalyst and the Data Revolution At the epicenter of this transformation is the Unified Payments Interface (UPI). Launched in 2016, UPI has evolved from a nascent payment rail into the undisputed backbone of the nation's retail economy. Its growth has been exponential, processing billions of transactions monthly. This platform has fundamentally rewired the economic behavior of hundreds of millions of citizens, from urban professionals to rural merchants. However, the true significance of UPI extends far beyond the convenience of instant, low-cost payments. Its most profound impact is the creation of a vast, high-velocity, and granular data stream. Every transaction—be it a utility bill payment, a grocery purchase, a peer-to-peer transfer, or a merchant settlement—leaves a permanent digital footprint. This "data exhaust" represents the most comprehensive, real-time ledger of financial behavior ever available in India. It is this data, rather than the payments themselves, that has become the single most valuable and disruptive asset for the financial services industry, particularly in the realm of credit.

1.3 The Enduring Problem of Financial Inclusion Despite this technological leap, a significant portion of the Indian population remains locked out of the formal credit ecosystem. Traditional credit underwriting, the bedrock of lending for decades, relies almost exclusively on data from credit information bureaus. These bureaus, in turn, build their scores based on an individual's past performance with formal credit products like loans and credit cards. This creates a persistent and well-documented paradox: to get credit, one must have a credit history, but to get a credit history, one must first get credit. This systemic barrier renders a vast segment of the population "credit invisible." This includes: "Thin-File" Individuals: Those with a limited or non-existent formal credit history. "New-to-Credit" (NTC) Individuals: Younger people or recent entrants to the workforce who have not yet had an opportunity to build a credit profile. Estimates place the number of Indians who fall into these categories in the hundreds of millions. These individuals, despite potentially having stable incomes and strong financial discipline (often managed through informal cash economies or, more recently, digital payments), are systematically overlooked or mispriced by legacy scoring models. This is not just a social challenge; it is a massive market failure, representing a vast, untapped economic opportunity for lenders who can find a reliable way to assess their creditworthiness.

1.4 The Emergence of Alternative Credit Scoring The data revolution powered by UPI provides a powerful solution to this problem. It allows for a paradigm shift from credit-based underwriting to cash-flow and behavior-based underwriting. Instead of asking, "How has this person paid back loans in the past?" lenders can now ask, "Does this person's real-time financial behavior demonstrate the capacity and willingness to repay a loan now?" This "alternative data" includes a wide array of non-traditional data points, with UPI transactions being the richest source. Academic and industry research has consistently shown that this data is highly predictive. The regularity of utility bill payments, the stability of monthly cash flows, and the patterns of spending are all powerful proxies for financial discipline and credit risk. By analyzing this digital footprint, lenders can finally "see" the credit invisible population and make data-driven decisions to extend credit.

1.5 Problem Statement and Research Gap The recognition of UPI data as a viable underwriting tool has led to a rapid proliferation of new credit scoring models. However, the market has not converged on a single solution. Instead, a review of the public-facing literature from fintechs, banks, and researchers reveals the emergence of three distinct and competing philosophies for how to best use this data: The Behavioral Scoring Model: Inspired by platforms like CARD91, this approach focuses on real-time, high-frequency patterns of transactions to detect acute risk. The Bank Statement Analysis (BSA) Model: Championed by

services like Precisa, this model aggregates historical transaction data to assess long-term financial stability. The Short-Term Rolling Features Model: A specialized model tied to new products like Credit Line on UPI (CLOU), this approach focuses on recent trends to manage revolving credit. While these models are frequently described in isolation, a significant gap exists in the literature: there is no single, consolidated, comparative analysis of these three philosophies. Lenders are faced with critical strategic questions: Is a real-time behavioral model more accurate than a stable historical one? How do the fairness risks and operational costs (like model drift and privacy compliance) differ between them? Which model is most appropriate for which credit product? This thesis aims to fill this critical research gap. It seeks to move beyond the description of individual models to build a comprehensive, multi-dimensional comparative framework.

1.6 Thesis Objectives and Scope

1.6.1 Objectives

The primary objective of this thesis is to systematically analyze and compare the three dominant UPI-centric credit scoring models to provide a strategic framework for financial institutions. This overarching objective is supported by the following sub-objectives: To systematically review and synthesize the public-domain literature (academic, industry, and news) to define and build a conceptual framework for each of the three scoring models. To conduct a multi-dimensional comparative analysis of these frameworks across four key criteria: Theoretical Predictive Power: The model's conceptual ability to accurately predict default. Model Explainability: The model's inherent transparency and compatibility with regulatory demands for interpretability. Algorithmic Fairness: The model's potential vulnerabilities to introducing or amplifying societal and demographic biases. Operational Viability: The practical challenges of implementation, including regulatory compliance, data infrastructure costs, and model maintenance (i.e., model drift). To synthesize these findings to provide clear, evidence-based strategic recommendations for lenders, researchers, and policymakers.

1.6.2 Scope and Delimitations

This thesis is a qualitative, literature-based review. The scope is strictly delimited to information and analysis derived from publicly available articles, news reports, industry whitepapers, and academic papers. This research does not involve the creation of new empirical models, the use of proprietary or non-public data, or quantitative benchmarking. The analysis is conceptual and strategic, aiming to structure the existing public knowledge into a novel, comparative framework to guide decision-making.

II. LITERATURE REVIEW

Literature Review and Problem Formulation

2.1 The State of the Art in Alternative Credit Scoring

The body of literature concerning credit scoring in India reveals a clear and consensus-driven narrative: the traditional, bureau-based scoring system is fundamentally insufficient for a nation with India's demographic and economic profile. This system, which relies on a history of formal loan repayments, is the primary cause of the "thin-file" problem, where hundreds of millions of creditworthy individuals are excluded from the formal credit ecosystem. In response, the literature identifies the rise of "alternative data" as the most viable solution. Research in this area, from both academic and industry sources, has two main streams. The first stream establishes the predictive power of various alternative data types. Studies consistently show that non-traditional financial data, such as the punctuality of utility bill payments and mobile phone top-ups, serves as a powerful proxy for an individual's financial discipline and is highly correlated with creditworthiness. The second, more recent stream focuses on the single most significant source of this alternative data: the Unified Payments Interface (UPI). The literature from financial technology and economic news sources uniformly identifies the high-velocity, high-volume data from UPI as a revolutionary asset for underwriting. This data provides a real-time, cash-flow-based view of an individual, a far more dynamic measure than a static, monthly bureau report. Concurrently, a parallel body of technical literature has explored the tools necessary to process this new, complex data. It is well-established that traditional linear models (like logistic regression, though still useful as a baseline) are inadequate for capturing the complex, non-linear patterns in high-dimensional behavioral data. Thus, the literature has shifted focus to machine learning (ML) algorithms. Numerous papers have benchmarked models like Random Forests and Gradient Boosting Machines (especially XGBoost),

demonstrating their superior performance in predictive accuracy and their ability to "discover" predictive signals that human analysts might miss.

2.2 Categorization of Emerging Scoring Philosophies

While the literature agrees on the problem (thin-file), the data (UPI), and the tool (ML), a systematic review reveals that it has not converged on a single solution. Instead, the public-domain information from fintech innovators, industry analysts, and researchers describes the emergence of three distinct, and at times competing, scoring philosophies. This thesis categorizes these as "The New Underwriting Trinity."

2.2.1 The Behavioral Scoring Model

This philosophy is primarily detailed in industry publications from new-age technology providers like CARD91. The core premise, as described, is that real-time patterns of behavior are more predictive of acute, near-term risk than historical aggregates. This model is event-driven, using AI and ML to analyze high-frequency UPI transaction streams. Key features cited in the literature include spending anomalies (e.g., sudden spikes or drops), transaction frequency, merchant category preferences, and the analysis of payment patterns (e.g., time-of-day, consistency) to generate a dynamic, often daily, risk score. A key strategic advantage highlighted is its ability to score individuals with no credit history whatsoever.

2.2.2 The Bank Statement Analysis (BSA) Model

This is an evolution of a more traditional underwriting technique, now automated and scaled using AI. As detailed by platforms like Precisa, this model is historical and aggregated. Its philosophy is that long-term financial stability and health are the most robust predictors of repayment. This model ingests and parses 6-12 months of transaction history, categorizing all inflows and outflows. The key features it extracts, as per the literature, are aggregated metrics like average monthly income, income stability, fixed expense-to-income ratios, cash flow volatility, and the detection of "red-flag" events like dishonored payments or suspicious circular transactions.

2.2.3 The Short-Term Rolling Features Model

This is the newest and most specialized philosophy, with its emergence tied directly to new credit products like the "Credit Line on UPI" (CLOU). Industry analyses from sources like Zeta and EY describe this model as recency-focused. Its purpose is not necessarily one-time underwriting but the dynamic, ongoing management of a revolving credit line. This model focuses on features calculated over a short, rolling window (e.g., the last 30-90 days). Key signals cited in the literature include recent transaction declines, bill payment punctuality, changes in merchant category risk (e.g., sudden spending at gambling sites), and credit utilization, all used to make real-time adjustments to a customer's available credit limit.

2.3 Thematic Analysis of Cross-Cutting Concepts

Beyond the specifics of these three models, the literature discusses a set of critical, cross-cutting concepts that apply to all forms of algorithmic underwriting.

2.3.1 Model Explainability (XAI)

A dominant theme, particularly in finance, is the "black-box" problem. While complex ML models (like XGBoost or deep neural networks) may be highly accurate, their decision-POST-making processes are often opaque. This presents a major challenge for a regulated industry that must be able to justify its lending decisions to auditors, regulators, and customers. The literature extensively discusses the use of "Explainable AI" (XAI) techniques, such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations), as essential tools to "look inside" these models and understand which features are driving risk, thereby balancing predictive power with transparency.

2.3.2 Algorithmic Fairness and Bias

A second, and perhaps more critical, theme is the risk of algorithmic bias. The literature warns that AI, if not carefully implemented, can learn, perpetuate, and even amplify existing societal biases. This is often not intentional but a product of "proxy discrimination," where a seemingly neutral feature (e.g., spending patterns, transaction locations, or even language used in app-based data) is highly correlated with a protected attribute like gender, socioeconomic status, or minority group. Academic and research papers in this domain are increasingly focused on methods for auditing models for fairness and implementing mitigation techniques to ensure that these new tools do not create new forms of digital discrimination.

2.3.3 Operational Viability: Drift and Regulation

Finally, the literature moves beyond the algorithm to discuss the practical realities of deployment. This theme is two-fold:

Model Drift: The UPI ecosystem and consumer behavior are not static; they are rapidly evolving. "Model drift" is a well-documented phenomenon where a model's predictive accuracy degrades over time because the real-world data patterns it was trained on have changed. This means a model is not a "set it and forget it" solution but a system that requires continuous monitoring and retraining.

Regulatory Compliance: The use of

personal financial data is strictly governed. The literature, especially from legal and compliance-focused sources, details the stringent framework set by the Reserve Bank of India (RBI) and the National Payments Corporation of India (NPCI) around data privacy, data minimization, and, most importantly, explicit user consent. These regulations place hard constraints on what data can be used, for how long, and for what purpose, acting as a critical filter on what is technologically possible.

2.4 Gaps in Existing Research

This systematic review of the literature reveals a significant and critical research gap. The existing body of knowledge is highly siloed. One set of sources (primarily industry) describes what the new models are (e.g., CARD91, Precisa). A second set (primarily academic) analyzes how to build components of them (e.g., "using XGBoost for scoring"). A third set (from legal and ethics) warns about the risks of doing so (e.g., "fairness in AI," "privacy"). What is definitively missing is the synthesis that connects these silos. No existing public-domain research provides a direct, multi-dimensional, comparative analysis of these three emerging scoring philosophies. There is no consolidated framework that places the Behavioral, BSA, and Rolling Features models side-by-side and evaluates them across a balanced scorecard of accuracy, explainability, fairness, and operational cost. This thesis directly addresses this gap. It provides the first "review of reviews" to synthesize these disparate threads into a single, cohesive framework. It moves beyond describing the individual pieces of the new lending landscape to analyzing the strategic trade-offs between them, thereby creating a novel and necessary guide for strategic decision-making in the new era of Indian digital credit.

III. RESEARCH METHODOLOGY

Proposed Methodology 3.1 Introduction

This chapter details the methodological framework designed to answer the research questions posed in Chapter 1. It serves as the bridge between the identified research gap (the lack of a comparative analysis of UPI-based scoring models, as established in Chapter 2) and the presentation of synthesized findings (in Chapter 4). Given that the underlying data, algorithms, and commercial strategies of the credit scoring models in question are proprietary and non-public, a direct empirical or quantitative study is not feasible. Therefore, this thesis adopts a qualitative, systematic literature review and comparative analysis as the most rigorous and appropriate methodology. This chapter outlines this approach in detail. It specifies the research design, the conceptual architecture of the analysis, the methods for data collection (literature sourcing) and analysis, the evaluation criteria used for the comparison, and the approach used to validate the findings.

3.2 Research Design

The research design for this thesis is a qualitative, descriptive, and comparative systematic review. This design was chosen for its specific suitability to a research problem where:

- The Subject is Emergent:** The field of UPI-based credit scoring is new, with practices and definitions still being established, making a survey of the existing landscape essential.
- Direct Data is Inaccessible:** The underlying financial transaction data and the proprietary scoring models are commercially sensitive and protected by privacy regulations, making primary empirical analysis impossible for an independent researcher.
- Public Knowledge is Abundant but Dispersed:** A vast amount of high-quality information exists in the public domain, but it is fragmented across academic papers, industry whitepapers, technical blogs, and regulatory reports. The objective of this design is not to create a new predictive model but to synthesize and structure the existing, dispersed public knowledge to reveal the strategic trade-offs, conceptual differences, and common challenges of the dominant scoring philosophies.

3.3 Conceptual and Analytical Architecture

The intellectual "system architecture" of this thesis is a multi-dimensional comparative framework. This framework forms the basis of the analysis and is constructed upon two primary pillars:

- The Subjects of Analysis (The "Trinity"):** These are the three distinct credit scoring philosophies identified in the literature review:
 - The Behavioral Scoring Model
 - The Bank Statement Analysis (BSA) Model
 - The Short-Term Rolling Features Model
- The Dimensions of Analysis (The "Evaluation Criteria"):** These are the four key strategic criteria, derived from the literature as being the most critical for any lending institution. These criteria serve as the "lenses" through which the three models are evaluated:
 - Theoretical Predictive Power
 - Model Explainability
 - Algorithmic Fairness
 - Operational Viability

The methodology consists of systematically

analyzing each of the three "subjects" (models) against each of the four "dimensions" (criteria). The output of this process is a comparative matrix of findings (presented in Chapter 4) that highlights the unique profile of strengths, weaknesses, and trade-offs for each model.

3.4 Data Collection and Analysis Methods

3.4.1 Data Collection (Literature Sourcing)

The "data" for this qualitative study consists of a curated corpus of publicly available documents. The collection process was systematic, involving:

- Source Identification:** Data was gathered from three primary types of public-domain sources to ensure a triangulated and balanced perspective:
- Academic Sources:** Peer-reviewed journals, conference proceedings (e.g., IEEE, arXiv), and university-hosted research papers.
- Industry Sources:** Whitepapers, technical blogs, and reports from fintech companies (e.g., CARD91, Precisa), technology providers (e.g., Zeta), and major consulting firms (e.g., EY).
- Media & Regulatory Sources:** Articles from reputable financial and technology news outlets (e.g., The Economic Times), and official publications from regulatory bodies (e.g., RBI, NPCI).

Search Strategy: A comprehensive search was conducted using keywords including: "UPI credit scoring," "alternative data India," "financial inclusion machine learning," "thin-file credit," "Bank Statement Analysis," "behavioral credit scoring," "Credit Line on UPI," "algorithmic fairness in lending," "model drift in finance," and "RBI digital lending guidelines."

Inclusion/Exclusion Criteria: Sources were included if they were published or updated post-2018 (to ensure relevance in the fast-evolving UPI landscape), written in English, and substantively discussed credit scoring within the Indian context. Sources were excluded if they were purely marketing materials, non-Indian, or lacked verifiable claims.

3.4.2 Data Analysis (Thematic Synthesis)

The collected body of literature was analyzed using a qualitative content analysis and thematic synthesis approach. This process involved:

- Coding and Categorization:** Each document was read and coded to identify key concepts, arguments, and evidence.
- Theme Identification:** These codes were then grouped into the analytical themes defined in Section 3.3 (i.e., the three models and the four criteria). For example, any mention of "real-time anomaly detection" was coded under "Behavioral Model" and "Predictive Power," while any mention of "proxy discrimination" was coded under "Algorithmic Fairness."
- Synthesis:** The final step involved synthesizing the coded information to build a coherent narrative and comparative argument for each model. This process focused on identifying points of consensus, disagreement, and unique trade-offs described across the different sources.

3.4.3 Purchasing Power Parity (PPP) & Cost-of-Living Normalization

To ensure that underwriting signals are comparable across India's heterogeneous price levels, we normalize all rupee-denominated cash-flow features by a regional price index. Let $PI_{r,t}$ denote the price index for region r at time t (PPP or sub-national cost-of-living proxy). For any nominal monetary feature XXX (income, expense, balance, EMI, etc.), we compute a real value:

Key normalized features used across models:

$$Income_{real} = \frac{Income}{PI_{r,t}}, \quad FixedExp_{real} = \frac{FixedExp}{PI_{r,t}}, \quad Discretionary_{real} = \frac{Discretionary}{PI_{r,t}}, \quad Affordability\ Margin_{real} = \frac{Affordability\ Margin}{PI_{r,t}},$$

$$AM_{real} = Income_{real} - FixedExp_{real}, \quad DSR_{real} = \frac{EMI_{real}}{Income_{real}}, \quad Volatility_{real} = \frac{CV\ of\ monthly\ Income_{real}}{Income_{real}}$$

Equivalized (optional): divide real amounts by a household equivalence scale if size is known. Why: (i) Improves cross-region comparability for BSA features, (ii) reduces proxy bias against lower-cost regions, and (iii) stabilizes rolling/behavioral thresholds where nominal amounts differ purely due to price levels.

Drift control: monitor $\Delta PI_{r,t}$ and retrain thresholds if cumulative change exceeds a governance limit (e.g., 5–10%).

3.5 Evaluation Criteria

As introduced in the analytical architecture, the comparison is based on four key criteria, which are justified as follows:

- Theoretical Predictive Power:** This criterion evaluates what the model predicts and how well it is suited for its specific task, as described in the literature. This moves beyond a generic claim of "accuracy" to assess its suitability for predicting acute vs. chronic risk.
- Model Explainability:** In a highly regulated field like finance, a model's accuracy is irrelevant if

its decisions are indefensible. This criterion, drawn from XAI research, evaluates the model's inherent transparency and its compatibility with regulatory demands for interpretability. Algorithmic Fairness: This criterion addresses the critical ethical and legal risk of algorithmic bias. Based on fairness-in-AI literature, it assesses each model's unique vulnerabilities to creating or amplifying demographic and societal biases. Operational Viability: This criterion provides a crucial real-world filter. It evaluates the practical challenges of deployment, including data privacy compliance (per RBI/NPCI rules), infrastructure costs, and long-term maintenance, with a specific focus on "model drift." This multi-dimensional scorecard ensures a holistic, strategic analysis that balances theoretical performance with the practical realities of business and regulation.

3.6 Validation Approach (Methodological Rigor and Triangulation) In a qualitative study, rigor and validity are not achieved through statistical testing but through the robustness of the analytical process. The validation approach for this thesis is triangulation. Triangulation, in this context, is the practice of cross-verifying findings by drawing from the different types of sources (academic, industry, and media). A claim or finding is considered "validated" or "robust" only if it is corroborated by multiple, independent source types. For example, if an industry whitepaper claims a feature is important for CLOU, an academic paper identifies a similar feature as predictive, and a regulatory report discusses rules for its use, the finding is considered triangulated and well-supported. This approach ensures that the thesis is not merely repeating the marketing claims of one company or the abstract theories of one academic paper. It produces a balanced, multi-perspective, and defensible analysis of the strategic landscape.

3.7 Conclusion This chapter has outlined the proposed methodology for the thesis. It employs a qualitative, systematic literature review structured around a multi-dimensional comparative framework. This design is the most rigorous and appropriate method for addressing the research question, given the impossibility of direct empirical access to proprietary models and data. The methodology ensures that the analysis is comprehensive, grounded in a diverse body of evidence, and structured to provide a clear, strategic comparison of the "New Underwriting Trinity." This defensible foundation supports the findings, analysis, and discussion presented in the subsequent chapter.

A. Purchasing Power Parity (PPP) & Cost-of-Living Normalization

3.4.3 Purchasing Power Parity (PPP) & Cost-of-Living Normalization To ensure that underwriting signals are comparable across India's heterogeneous price levels, we normalize all rupee-denominated cash-flow features by a regional price index. Let $PI_{r,t}$ denote the price index for region r at time t (PPP or sub-national cost-of-living proxy). For any nominal monetary feature XXX (income, expense, balance, EMI, etc.), we compute a real value: Key normalized features used across models:

$$Income_{real} = \frac{Income}{PI_{r,t}}$$

$$FixedExp_{real} = \frac{FixedExp}{PI_{r,t}}$$

$$Discretionary_{real} = \frac{Discretionary}{PI_{r,t}}$$

Affordability Margin (real):

$$AM_{real} = \frac{Income_{real} - FixedExp_{real}}{Income_{real}}$$

DSR (real):

$$DSR_{real} = \frac{EMI_{real}}{Income_{real}}$$

Volatility (real):

$$CV = \frac{SD}{Mean}$$

CV of monthly $Income_{real}$ and AM_{real}

Equivalized (optional): divide real amounts by a household equivalence scale if size is known. Why: (i) Improves cross-region comparability for BSA features, (ii) reduces proxy bias against lower-cost regions, and (iii) stabilizes rolling/behavioral thresholds where nominal amounts differ purely due to price levels. Drift control: monitor $\Delta PI_{r,t}$ and retrain thresholds if cumulative change exceeds a governance limit (e.g., 5–10%).

IV. RESULTS

Results and Discussion 4.1 Introduction to Synthesized Results This chapter presents the primary results of the thesis, derived from the systematic literature review and comparative analysis methodology detailed in Chapter 3. The "results" in this qualitative study are not new empirical data but rather the synthesized findings from the public domain literature, structured systematically to answer the research question. The chapter is organized as follows: Sections 4.2, 4.3, and 4.4 present the detailed findings for each of the three identified credit scoring models—the "New Underwriting Trinity"—evaluated against the four key criteria: theoretical predictive power, model explainability, algorithmic fairness, and operational viability. Section

V. DISCUSSION

4.5 then moves from results to a holistic discussion, providing a comparative analysis of the models and synthesizing the core strategic trade-offs that emerge from the findings.

4.2 Detailed Findings: The Behavioral Scoring Model The Behavioral Scoring model, as conceptualized in the literature from sources like CARD91 and NASSCOM , is an event-driven, real-time approach. The findings regarding this model are as follows:

Predictive Power: The literature posits that this model's primary strength is its high predictive accuracy in identifying acute, near-term risk. Its value is not in assessing long-term stability but in detecting imminent changes in a borrower's financial state. It does this by analyzing high-frequency data for patterns and anomalies. Key predictive features cited include spending velocity, payment failures, shifts in merchant category spending, and consistency of inflows/outflows. Paired with complex machine learning algorithms, it can capture subtle, non-linear signals that precede a default.

Model Explainability: This is a significant area of weakness. Because the model's power is derived from complex, non-linear interactions discovered by algorithms like XGBoost , it is inherently a "black box." The findings show that this model has low intrinsic interpretability. It is therefore heavily, if not completely, reliant on post-hoc Explainable AI (XAI) techniques like SHAP and LIME to provide justifications for its decisions, a necessary step for regulatory compliance.

Algorithmic Fairness: The literature on fairness in AI highlights significant vulnerabilities for this model. The findings indicate a high risk of "proxy discrimination." The model's reliance on "pattern-of-life" analysis means it could learn to penalize non-traditional or "anomalous" spending behaviors that may be common, but not risky, for certain demographic or socioeconomic groups. It may, for example, incorrectly flag a user with erratic income (like a gig worker) or unconventional spending (like a rural merchant) as high-risk.

Operational Viability: This is the model's most significant hurdle.

Privacy & Regulation: It requires access to the most granular, real-time, and continuous stream of a user's transaction data. This creates a high compliance burden under RBI and NPCI mandates for explicit, specific, and revocable consent, posing significant data privacy risks.

Model Drift: The findings show this model is the most vulnerable to model drift. Consumer behavior patterns are not static; they evolve rapidly. A model trained on pre-pandemic spending patterns, for example, would be useless today. This necessitates a costly, continuous MLOps (Machine Learning Operations) pipeline for monitoring, re-training, and re-deploying the model.

Infrastructure: It requires a robust, low-latency, real-time data processing and decisioning infrastructure, which is a substantial technical and financial investment.

4.3 Detailed Findings: The Bank Statement Analysis (BSA) Model The BSA model, as described by platforms like Precisa , is an evolution of traditional underwriting. It is a historical, aggregate-based approach.

Predictive Power: The literature presents this model's strength not as acuity, but as robustness and stability. By aggregating 6-12 months of transaction history, it excels at establishing a borrower's foundational financial health. Key predictive features are aggregated metrics such as average monthly income, income stability (e.g., coefficient of variation), recurring fixed-expense ratios, and the identification of major negative events (e.g., check bounces, low-balance days). It provides a strong, stable baseline of creditworthiness but is, by design, slow to react to sudden, real-time changes in a borrower's status.

Model Explainability: This is the model's greatest strength. The findings show it has the highest intrinsic interpretability of the three. Its features (e.g., "Average Monthly Income," "Rent-to-Income Ratio") are based on established, intuitive financial principles that are easily understood and explained to both regulators and customers, minimizing the "black-box"

problem. **Algorithmic Fairness:** While more interpretable, the BSA model is not immune to bias. The literature suggests its primary vulnerability lies in its potential to penalize income volatility. A model trained on traditional, salaried employees may unfairly downgrade gig economy workers, freelancers, or small business owners who have a high but non-linear income, thereby creating a systemic bias against non-traditional employment. **Operational Viability:** This model is found to be the most operationally straightforward. **Privacy & Regulation:** The consent request is clear and time-bound ("provide your last 6 months of bank statements"). This is a one-time data pull, which is simpler from a compliance perspective than continuous monitoring. **Model Drift:** It is the most stable and least vulnerable to drift. The fundamental principles of financial health (e.g., "income should exceed expenses") are far more stable over time than specific behavioral spending patterns. **Infrastructure:** It can be run in batch processes (e.g., during an application), requiring less complex and costly real-time infrastructure.

4.4 Detailed Findings: The Short-Term Rolling Features Model This is the newest and most specialized model, with its emergence tied directly to new products like the Credit Line on UPI (CLOU), as detailed by sources like Zeta and EY. **Predictive Power:** The findings are clear that this model's power is highly specialized. It is not designed for one-time underwriting of new-to-credit customers. Its purpose is the dynamic portfolio management of existing customers with a revolving credit line. It focuses on a recency-weighted, rolling window (e.g., the last 30-90 days) of behavior. Key predictive features are high-stakes, recent events: success or failure of recent bill payments, changes in credit utilization, and recent transactions with high-risk merchant categories. **Model Explainability:** Like the Behavioral model, this model is complex, pattern-based, and often non-linear, giving it low intrinsic interpretability. It is heavily reliant on XAI tools to explain its decisions, which is critical in this context (e.g., "Why was my credit limit suddenly reduced?"). **Algorithmic Fairness:** Its fairness risks are similar to the Behavioral model but are concentrated on recency. For example, a model that penalizes a recent, short-term dip in income (e.g., due to illness) could create a "downward spiral" for the user, reducing their credit limit just when they need it most. **Operational Viability:** This model has high operational complexity. **Privacy & Regulation:** It implies continuous or frequent monitoring of an existing customer's transactions, which requires a clear, ongoing consent model that must be carefully managed. **Drift:** It is highly vulnerable to drift, as the short-term patterns that predict default in a revolving portfolio can change rapidly with the macroeconomic environment. **Infrastructure:** It requires a robust system for ingesting data, re-scoring customers frequently (potentially daily), and pushing new credit limits to the core banking system in a highly automated loop.

4.5 Comparative Discussion and Analysis The synthesized findings from the literature reveal that there is no single "best" credit scoring model. The choice represents a series of critical, multi-dimensional trade-offs. The "New Underwriting Trinity" does not present three interchangeable options, but three distinct tools for different strategic purposes. The core trade-off that emerges from the analysis is a trilemma between Predictive Acuity, Interpretability, and Operational Cost. The Behavioral Model maximizes predictive acuity for acute risk but does so at the expense of high operational cost and low interpretability. The BSA Model maximizes interpretability and low operational cost but does so by sacrificing acute predictive acuity (favoring long-term stability). The Rolling Features Model is a specialized tool that provides high predictive acuity for portfolio management, but, like the Behavioral model, comes with high operational cost and low interpretability. The literature, when synthesized, points toward a logical, hybrid solution. A sophisticated lender would not choose one model but would likely build a "stack" that combines their strengths: **For Underwriting:** The BSA Model would be used as the foundational layer for one-time underwriting of a new customer. Its stability, robustness, and high interpretability make it ideal for establishing a baseline of creditworthiness and setting an initial credit limit in a regulator-friendly manner. **For Portfolio Management & Fraud:** The Behavioral/Rolling Features Model would be used as a real-time, "over-the-top" monitoring system for existing customers. This layer would manage dynamic fraud detection, identify acute distress signals (to trigger proactive outreach), and dynamically manage the limits of revolving credit products like CLOU. This hybrid approach, which is implicitly suggested by the different focuses of the literature, represents a mature strategy. It balances the stable, robust, and interpretable nature of the BSA model with the high-acuity, real-time risk detection of the behavioral and rolling-feature models.

VI. CONCLUSION AND FUTURE WORK

Conclusions and Scope for Future Work

5.1 Conclusions This thesis conducted a systematic literature review to build a comprehensive, multi-dimensional comparative framework for the three dominant UPI-centric credit scoring philosophies in India. It synthesized disparate public information to analyze the Behavioral Scoring Model, the Bank Statement Analysis (BSA) Model, and the Short-Term Rolling Features Model across the dimensions of predictive power, explainability, fairness, and operational viability. The research concludes that the emergence of UPI data has fundamentally changed the underwriting landscape, but it has not produced a single silver-bullet solution. Instead, it has created a "New Underwriting Trinity," with each model serving a distinct purpose and carrying an innate profile of trade-offs. The principal conclusions of this research are: There is no universally "best" model. The choice of a scoring model is not a technical decision but a core strategic one, defined by a series of trade-offs. The primary trade-off is a trilemma between predictive acuity, model interpretability, and operational cost (including compliance and maintenance). The Behavioral Model offers the highest potential for acute risk detection but at the high cost of low interpretability, high privacy risk, and high vulnerability to model drift. The BSA Model serves as a robust and stable foundation. It maximizes interpretability and operational stability and is well-suited for baseline underwriting, though it sacrifices the ability to detect real-time changes in risk. The Rolling Features Model is a specialized, high-maintenance tool, not for general underwriting, but for the dynamic portfolio management of revolving credit products, where its recency-focus is a key strength. All three models are vulnerable to algorithmic bias in distinct ways. The Behavioral model risks penalizing non-traditional patterns, while the BSA model risks penalizing income volatility, highlighting that fairness must be a conscious design choice, not an assumed outcome. The most mature and strategically sound approach, as suggested by the synthesized literature, is a hybrid model. This approach would use the stable, interpretable BSA model for one-time underwriting and setting initial credit limits, while employing a real-time Behavioral or Rolling Features model as an "over-the-top" system for ongoing fraud detection, acute risk monitoring, and the dynamic management of revolving credit lines.

5.2 Scope for Future Work This thesis, being a qualitative and literature-based review, provides a conceptual and strategic framework. Its primary limitation is the absence of empirical, quantitative data. This limitation, however, directly illuminates the most critical areas for future research:

Quantitative Benchmarking: The most significant gap is the lack of public, empirical studies that quantify the trade-offs discussed in this thesis. Future research using privacy-preserving synthetic datasets or proxy data would be invaluable to measure the exact performance lift (e.g., in AUC or Gini coefficient) of each model.

Bias Mitigation in the Indian Context: The literature on algorithmic fairness is predominantly based on US or European datasets. Research is urgently needed to develop and test bias-mitigation strategies that are specifically tailored to the unique demographic and social constructs of India.

Long-Term Model Drift Analysis: The UPI ecosystem is still new and evolving. A longitudinal study that tracks the performance decay (model drift) of these different model architectures over several years would provide critical insights into their long-term maintenance costs and reliability.

Feasibility of Privacy-Enhancing Technologies (PETs): Research into the application of PETs, such as federated learning or differential privacy, could explore how lenders might gain the predictive benefits of granular behavioral data without creating the systemic privacy risks discussed in this thesis.

Integration of Purchasing Power Parity (PPP) and Regional Cost-of-Living Normalization: The literature reviewed in this thesis does not incorporate geographic cost-of-living differences or purchasing power variations into cash-flow based credit scoring models. However, economic research (World Bank ICP; RBI Regional CPI Bulletins) consistently demonstrates that the real purchasing value of income and expenses varies significantly across Indian states and urban–rural segments. For instance, the same nominal monthly income may indicate substantially different affordability and repayment capacity depending on the local price level. Future work should therefore explore the integration of PPP-adjusted or regional price index-normalized financial features into UPI-based underwriting and risk monitoring models. Such normalization has the potential to reduce geographic proxy bias and improve fairness, particularly for borrowers in lower-cost regions who may currently be misclassified when models rely only on nominal values.

Empirical evaluation is recommended to measure the effect of PPP adjustments on model stability, segmentation accuracy, and inclusion outcomes across diverse socio-economic geographies.

5.3 Final Remarks

The intersection of artificial intelligence, high-volume digital data, and the mission of financial inclusion in India represents one of the most significant socioeconomic experiments of our time. The "New Underwriting Trinity" analyzed in this thesis provides the toolkit for this experiment, holding the potential to extend formal credit to hundreds of millions of deserving individuals. However, this potential is balanced by significant risks. The tools of inclusion, if implemented without care, can become tools of exclusion, creating new, algorithm-driven barriers that are faster, more opaque, and more scalable than the old ones. The findings of this thesis serve as a structured guide for navigating this new landscape. They argue for a strategic, hybrid, and clear-eyed approach—one that balances the pursuit of predictive accuracy with the non-negotiable demands of interpretability, fairness, and regulatory compliance. The ultimate success of this new digital credit ecosystem will be measured not just by its profitability, but by its ability to build a system that is not only more inclusive, but also more stable and just.

REFERENCES

- S. Agarwal, S. Alok, P. Ghosh, and S. Gupta, Financial Inclusion and Alternate Credit Scoring for the Millennials: Role of Big Data and Machine Learning in Fintech. New York, NY, USA: NYU Stern School of Business, 2020.
- D. Shukla and S. Gupta, Machine Learning Algorithms for Credit Scoring of Thin-File Consumers. Jaipur, India: Jaipur National University, 2025.
- CARD91, "CARD91 Launches AI-Powered UPI Credit Scoring Engine to Enable Smarter Lending and Drive Financial Inclusion," PR Newswire, 2025. [Online]. Precisa, "Alternative Credit Scoring Using Bank Statement Analysis," Precisa Blog, 2025. [Online]. Zeta, "Credit Line on UPI: Why Proactive Planning is Key to Success," Zeta Blog, 2024. [Online]. Ernst & Young, "How recent changes to UPI are helping fill India's credit gap," 2023. [Online]. Kumar, et al., "Credit Scoring System Using Machine Learning," Int. J. Sci. Res. Comput. Sci. Eng. Inf. Technol., 2024.
- R. Hlongwane, K. Ramaboa, and W. Mongwe, "Enhancing credit scoring accuracy with a comprehensive evaluation of alternative data," PLOS ONE, vol. 19, no. 1, p. e0297008, Jan. 2024.
- Fuster, P. Goldsmith-Pinkham, T. Ramadorai, and A. Walther, "The Fairness of Credit Scoring Models," arXiv:2202.08833, Feb. 2022.
- P. Wilson, Explainable AI in Finance. CFA Institute Research Foundation, 2025.
- B. HadjiMisheva, D. Gligorijevikj, and H. S. B. E. G. D. G. S. E. D. M. G. M. S. S. A. G. Gjorgjevikj, "SHAP and LIME: An Evaluation of Discriminative Power in Credit Risk," Front. Artif. Intell., vol. 4, Jul. 2021.
- World Bank, The Use of Alternative Data in Credit Risk Assessment - Opportunities, Risks, and Challenges. Washington, D.C., USA: World Bank Group, 2025.
- K. Sinha, "Digital lending regulation landscape," Law.asia, 2025. [Online]. Deepchecks, "Model Drift: How it Affects Your Predictive Models and What to Do About It," Deepchecks Blog, 2024. [Online]. NASSCOM Community, "AI-Powered Credit Scoring Using Alternative Data: Redefining Financial Inclusion," 2024. [Online]. The Economic Times, "NPCI eyes EMI payments on UPI in fresh credit play push," 2025. [Online]. P. Mungara, et al., "A Study of UPI User Security & Privacy in India," in Proc. 34th USENIX Security Symposium, 2025.
- D. Sharma, Improving Credit Scoring with Random Forests. Edinburgh, UK: University of Edinburgh Business School, 2023.
- H. He, Z. Wang, and Z. Jiang, "Improved Credit Scoring Model with Hyperparameter Optimization," Springer Professional, 2024.
- Credit-Scoring.co.uk, "Fairness and Bias in Credit Risk Models," 2024. [Online]. J. P. Singh, "Failure to Launch for the Indian Credit Invisible: An Analysis of the 'Thin-File' Problem," IIM Bangalore Management Review, vol. 35, no. 1, pp. 22-34, Mar. 2023.
- S. C. Sharma, "Credit Line on UPI (CLOU): The Next Frontier in Digital Lending," The Journal of Digital Banking, vol. 8, no. 1, pp. 45-59, 2024.
- R. Kumar and V. Sharma, "Monitoring and Managing Concept Drift in Financial Credit Scoring Models," IEEE Trans. Knowl. Data Eng., vol. 35, no. 2, pp. 1123-1135, Feb. 2023.
- Reserve Bank of India, "Guidelines on Digital Lending," RBI/2022-23/111, Mumbai, India, Sep. 2022. [Online]. NITI Aayog, "Fintech- The Force of India's Trillion-Dollar Digital Economy," NITI Aayog, New Delhi, India, 2024. [Online].